

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$\frac{3}{(2x-1)(x+1)} = \frac{A}{2x-1} + \frac{B}{x+1} \Rightarrow A = \dots, B = \dots$	M1	1.1b
	Either $A = 2$ or $B = -1$	A1	1.1b
	$\frac{3}{(2x-1)(x+1)} = \frac{2}{2x-1} - \frac{1}{x+1}$	A1	1.1b
	(3)		
(b)	$\int \frac{1}{V} dV = \int \frac{3}{(2t-1)(t+1)} dt$	B1	1.1a
	$\int \frac{2}{2t-1} - \frac{1}{t+1} dt = \dots \ln(2t-1) - \dots \ln(t+1) (+c)$	M1	3.1a
	$\ln V = \ln(2t-1) - \ln(t+1) (+c)$	A1ft	1.1b
	Substitutes $t = 2, V = 3 \Rightarrow c = (\ln 3)$	M1	3.4
	$\ln V = \ln(2t-1) - \ln(t+1) + \ln 3$ $V = \frac{3(2t-1)}{(t+1)} *$	A1*	2.1
	(5)		

	(b) Alternative separation of variables:		
	$\int \frac{1}{3V} dV = \int \frac{1}{(2t-1)(t+1)} dt$	B1	1.1a
	$\frac{1}{3} \int \frac{2}{2t-1} - \frac{1}{t+1} dt = \dots \ln(2t-1) - \dots \ln(t+1) (+c)$	M1	3.1a
	$\frac{1}{3} \ln 3V = \frac{1}{3} \ln(2t-1) - \frac{1}{3} \ln(t+1) (+c)$	A1ft	1.1b
	Substitutes $t = 2, V = 3 \Rightarrow c = \left(\frac{1}{3} \ln 3\right)$	M1	3.4
	$\frac{1}{3} \ln V = \frac{1}{3} \ln(2t-1) - \frac{1}{3} \ln(t+1) + \frac{1}{3} \ln 3$ $V = \frac{3(2t-1)}{(t+1)} *$	A1*	2.1
	(5)		
(c)	(i) 30 (minutes)	B1	3.2a
	(ii) 6 (m ³)	B1	3.4
	(2)		
(10 marks)			
Notes:			

(a)

M1: Correct method of partial fractions leading to values for their A and B

E.g. substitution: $\frac{3}{(2x-1)(x+1)} = \frac{A}{2x-1} + \frac{B}{x+1} \Rightarrow 3 = A(x+1) + B(2x-1) \Rightarrow A = \dots, B = \dots$

Or compare coefficients $\frac{3}{(2x-1)(x+1)} = \frac{A}{2x-1} + \frac{B}{x+1} \Rightarrow 3 = x(A+2B) + A - B \Rightarrow A = \dots, B = \dots$

Note that $\frac{3}{(2x-1)(x+1)} = \frac{A}{2x-1} + \frac{B}{x+1} \Rightarrow 3 = A(2x-1) + B(x+1) \Rightarrow A = \dots, B = \dots$ scores M0



A1: Correct value for "A" or "B"

A1: Correct partial fractions not just values for "A" and "B". $\frac{2}{2x-1} - \frac{1}{x+1}$ or e.g. $\frac{2}{2x-1} + \frac{-1}{x+1}$

Must be seen as fractions but if not stated here, allow if the correct fractions appear later.

(b)

B1: Separates variables $\int \frac{1}{V} dV = \int \frac{3}{(2t-1)(t+1)} dt$. May be implied by later work.

Condone omission of the integral signs but the dV and dt must be in the correct positions if awarding this mark in isolation but they may be implied by subsequent work.

M1: Correct attempt at integration of the partial fractions.

Look for $\dots \ln(2t-1) + \dots \ln(t+1)$ where $\dots$ are constants.

Condone missing brackets around the $(2t-1)$ and/or the $(t+1)$ for this mark

Alft: Fully correct equation following through their A and B only.

No requirement for $+c$ here.

The brackets around the $(2t-1)$ and/or the $(t+1)$ must be seen or implied for this mark

M1: Attempts to find "c" or e.g. " $\ln k$ " using $t=2$, $V=3$ following an attempt at integration.

Condone poor algebra as long as $t=2$, $V=3$ is used to find a value of their constant.

Note that the constant may be found immediately after integrating or e.g. after the $\ln$'s have been combined.

A1*: Correct processing leading to the given answer $V = \frac{3(2t-1)}{(t+1)}$

Alternative:

B1: Separates variables $\int \frac{1}{3V} dV = \int \frac{1}{(2t-1)(t+1)} dt$. May be implied by later work.

Condone omission of the integral signs but the dV and dt must be in the correct positions if awarding this mark in isolation but they may be implied by subsequent work.

M1: Correct attempt at integration of the partial fractions.

Look for $\dots \ln(2t-1) + \dots \ln(t+1)$ where $\dots$ are constants.

Condone missing brackets around the $(2t-1)$ and/or the $(t+1)$ for this mark

Alft: Fully correct equation following through their A and B only.

No requirement for $+c$ here.

The brackets around the $(2t-1)$ and/or the $(t+1)$ must be seen or implied for this mark

M1: Attempts to find "c" or e.g. " $\ln k$ " using $t=2$, $V=3$ following an attempt at integration.

Condone poor algebra as long as $t=2$, $V=3$ is used to find a value of their constant.

Note that the constant may be found immediately after integrating or e.g. after the $\ln$'s have been combined.

A1*: Correct processing leading to the given answer $V = \frac{3(2t-1)}{(t+1)}$

(Note the working may look like this:

$$\frac{1}{3} \ln 3V = \frac{1}{3} \ln(2t-1) - \frac{1}{3} \ln(t+1) + c, \quad \frac{1}{3} \ln 9 = \frac{1}{3} \ln(3) - \frac{1}{3} \ln 3 + c, \quad c = \frac{1}{3} \ln 9$$

$$\ln 3V = \ln \frac{9(2t-1)}{(t+1)} \Rightarrow 3V = \frac{9(2t-1)}{(t+1)} \Rightarrow V = \frac{3(2t-1)}{(t+1)} *$$

Note that B0M1A1M1A1 is not possible in (b) as the B1 must be implied if all the other marks have been awarded.

Note also that some candidates may use different variables in (b) e.g.

$$\frac{dy}{dx} = \frac{3y}{(2x-1)(x+1)} \Rightarrow \int \frac{1}{y} dy = \int \frac{3}{(2x-1)(x+1)} dx \text{ etc. In such cases you should award marks for}$$

equivalent work but they must revert to the given variables at the end to score the final mark.

Also if e.g. a "t" becomes an "x" within their working but is recovered allow full marks.

(c)

B1: Deduces 30 minutes. Units not required so just look for 30 but allow equivalents e.g. $\frac{1}{2}$ an hour.

If units are given they must be correct so do not allow e.g. 30 hours.

B1: Deduces 6 m^3 . Units not required so just look for 6. Condone $V < 6$ or $V \leq 6$

If units are given they must be correct so do not allow e.g. 6 m.

Q2.

Question	Scheme	Marks	AOs
	$\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv A + \frac{B}{(x-3)} + \frac{C}{(1-2x)}$		
(a) Way 1	$1+11x-6x^2 \equiv A(1-2x)(x-3) + B(1-2x) + C(x-3) \Rightarrow B = \dots, C = \dots$	M1	2.1
	$A = 3$	B1	1.1b
	Uses substitution or compares terms to find either $B = \dots$ or $C = \dots$	M1	1.1b
	$B = 4$ and $C = -2$ which have been found using a correct identity	A1	1.1b
	(4)		
(a) Way 2	{long division gives} $\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv 3 + \frac{-10x+10}{(x-3)(1-2x)}$		
	$-10x+10 \equiv B(1-2x) + C(x-3) \Rightarrow B = \dots, C = \dots$	M1	2.1
	$A = 3$	B1	1.1b
	Uses substitution or compares terms to find either $B = \dots$ or $C = \dots$	M1	1.1b
	$B = 4$ and $C = -2$ which have been found using $-10x+10 \equiv B(1-2x) + C(x-3)$	A1	1.1b
	(4)		
(b)	$f(x) = 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)} \quad \{ = 3 + 4(x-3)^{-1} - 2(1-2x)^{-1} \}; x > 3$		
	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2} \left\{ = -\frac{4}{(x-3)^2} - \frac{4}{(1-2x)^2} \right\}$	M1 A1ft	2.1 1.1b
	Correct $f'(x)$ and as $(x-3)^2 > 0$ and $(1-2x)^2 > 0$, then $f'(x) = -(+ve) - (+ve) < 0$, so $f(x)$ is a decreasing function	A1	2.4
	(3)		
(7 marks)			

Notes for Question	
(a)	
M1:	Way 1: Uses a correct identity $1+11x-6x^2 \equiv A(1-2x)(x-3)+B(1-2x)+C(x-3)$ in a complete method to find values for B and C . Note: Allow one slip in copying $1+11x-6x^2$ Way 2: Uses a correct identity $-10x+10 \equiv B(1-2x)+C(x-3)$ (which has been found from long division) in a complete method to find values for B and C
B1:	$A=3$
M1:	Attempts to find the value of either B or C from their identity This can be achieved by <i>either</i> substituting values into their identity <i>or</i> by comparing coefficients and solving the resulting equations simultaneously
A1:	See scheme
Note:	Way 1: Comparing terms: $x^2: -6 = -2A; x: 11 = 7A - 2B + C; \text{ constant: } 1 = -3A + B - 3C$ Way 1: Substituting: $x=3: -20 = -5B \Rightarrow B=4; x=\frac{1}{2}: 5 = -\frac{5}{2}C \Rightarrow C=-2$
Note:	Way 2: Comparing terms: $x: -10 = -2B + C; \text{ constant: } 10 = B - 3C$ Way 2: Substituting: $x=3: -20 = -5B \Rightarrow B=4; x=\frac{1}{2}: 5 = -\frac{5}{2}C \Rightarrow C=-2$
Note:	$A=3, B=4, C=-2$ from no working scores M1B1M1A1
Note:	The final A1 mark is effectively dependent upon both M marks

Notes for Question Continued	
(a) ctd	
Note:	Writing $1+11x-6x^2 \equiv B(1-2x)+C(x-3) \Rightarrow B=4, C=-2$ will get 1 st M0, 2 nd M1, 1 st A0
Note:	Way 1: You can imply a correct identity $1+11x-6x^2 \equiv A(1-2x)(x-3)+B(1-2x)+C(x-3)$ from seeing $\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{A(1-2x)(x-3)+B(1-2x)+C(x-3)}{(x-3)(1-2x)}$
Note:	Way 2: You can imply a correct identity $-10x+10 \equiv B(1-2x)+C(x-3)$ from seeing $\frac{-10x+10}{(x-3)(1-2x)} \equiv \frac{B(1-2x)+C(x-3)}{(x-3)(1-2x)}$
(b)	
M1:	Differentiates to give $\{f'(x) = \} \pm \lambda(x-3)^{-2} \pm \mu(1-2x)^{-2}; \lambda, \mu \neq 0$
A1ft:	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2}$, which can be simplified or un-simplified
Note:	Allow A1ft for $f'(x) = -(\text{their } B)(x-3)^{-2} + (2)(\text{their } C)(1-2x)^{-2}; (\text{their } B), (\text{their } C) \neq 0$
A1:	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2}$ or $f'(x) = -\frac{4}{(x-3)^2} - \frac{4}{(1-2x)^2}$ and a correct explanation e.g. $f'(x) = -(+ve) - (+ve) < 0$, so $f(x)$ is a decreasing {function}
Note:	The final A mark can be scored in part (b) from an incorrect $A = \dots$ or from $A = 0$ or no value of A found in part (a)

Notes for Question Continued - Alternatives						
(a)						
Note:	Be aware of the following alternative solutions, by initially dividing by " $(x-3)$ " or " $(1-2x)$ "					
	$\bullet \frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{-6x-7}{(1-2x)} - \frac{20}{(x-3)(1-2x)} \equiv 3 - \frac{10}{(1-2x)} - \frac{20}{(x-3)(1-2x)}$ $\frac{20}{(x-3)(1-2x)} \equiv \frac{D}{(x-3)} + \frac{E}{(1-2x)} \Rightarrow 20 \equiv D(1-2x) + E(x-3) \Rightarrow D = -4, E = -8$ $\Rightarrow 3 - \frac{10}{(1-2x)} - \left(\frac{-4}{(x-3)} + \frac{-8}{(1-2x)} \right) \equiv 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)}; A=3, B=4, C=-2$ $\bullet \frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{3x-4}{(x-3)} + \frac{5}{(x-3)(1-2x)} \equiv 3 + \frac{5}{(x-3)} + \frac{5}{(x-3)(1-2x)}$ $\frac{5}{(x-3)(1-2x)} \equiv \frac{D}{(x-3)} + \frac{E}{(1-2x)} \Rightarrow 5 \equiv D(1-2x) + E(x-3) \Rightarrow D = -1, E = -2$ $\Rightarrow 3 + \frac{5}{(x-3)} + \left(\frac{-1}{(x-3)} + \frac{-2}{(1-2x)} \right) \equiv 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)}; A=3, B=4, C=-2$					
(b)						
	Alternative Method 1:					
	$f(x) = \frac{1+11x-6x^2}{(x-3)(1-2x)}, x > 3 \Rightarrow f(x) = \frac{1+11x-6x^2}{-2x^2+7x-3}; \begin{cases} u = 1+11x-6x^2 & v = -2x^2+7x-3 \\ u' = 11-12x & v' = -4x+7 \end{cases}$					
	$f'(x) = \frac{(-2x^2+7x-3)(11-12x) - (1+11x-6x^2)(-4x+7)}{(-2x^2+7x-3)^2}$	<table><tr><td>Uses quotient rule to find $f'(x)$</td><td>M1</td></tr><tr><td>Correct differentiation</td><td>A1</td></tr></table>	Uses quotient rule to find $f'(x)$	M1	Correct differentiation	A1
Uses quotient rule to find $f'(x)$	M1					
Correct differentiation	A1					
	$f'(x) = \frac{-20((x-1)^2+1)}{(-2x^2+7x-3)^2}$ and a correct explanation, e.g. $f'(x) = -\frac{(+ve)}{(+ve)} < 0$, so $f(x)$ is a decreasing {function}	A1				
	Alternative Method 2:					
	Allow M1A1A1 for the following solution:					
	Given $f(x) = 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)} = 3 + \frac{4}{(x-3)} + \frac{2}{(2x-1)}$					
	as $\frac{4}{(x-3)}$ decreases when $x > 3$ and $\frac{2}{(2x-1)}$ decreases when $x > 3$					
	then $f(x)$ is a decreasing {function}					

Q3.

Question	Scheme	Marks	AOs
(a)	Attempts to solve $\frac{3}{2} = \frac{8x+5}{2x+3} \Rightarrow x = \dots$	M1	3.1a
	Or substitutes $x = \frac{3}{2}$ into $\frac{5-3x}{2x-8}$		
	$\left(f^{-1}\left(\frac{3}{2}\right)\right) = -\frac{1}{10}$	A1	1.1b
	(2)		
(b)	$\left(\frac{8x+5}{2x+3}\right) = 4 \pm \frac{\dots}{2x+3}$	M1	1.1b
	$\left(\frac{8x+5}{2x+3}\right) = 4 - \frac{7}{2x+3}$	A1	2.1
	(2)		
(c)	$0 \leq g^{-1}(x) \leq 4$	B1	2.2a
	(1)		
(d)	Attempts either boundary $f(0) = \frac{8 \times 0 + 5}{2 \times 0 + 3}$ or $f(4) = \frac{8 \times 4 + 5}{2 \times 4 + 3}$	M1	3.1a
	Attempts both boundaries $f(0) = \frac{8 \times 0 + 5}{2 \times 0 + 3}$ and $f(4) = \frac{8 \times 4 + 5}{2 \times 4 + 3}$	dM1	1.1b
	Range $\frac{5}{3} \leq fg^{-1}(x) \leq \frac{37}{11}$	A1	2.1
	(3)		
	Alternative by attempting $fg^{-1}(x)$		
	$g^{-1}(x) = \sqrt{16-x} \Rightarrow fg^{-1}(x) = \frac{8\sqrt{16-x}+5}{2\sqrt{16-x}+3}$	M1	3.1a
	$fg^{-1}(0) = \frac{8 \times 4 + 5}{2 \times 4 + 3}$ or $fg^{-1}(16) = \frac{8 \times 0 + 5}{2 \times 0 + 3}$		
	$fg^{-1}(0) = \frac{8 \times 4 + 5}{2 \times 4 + 3}$ and $fg^{-1}(16) = \frac{8 \times 0 + 5}{2 \times 0 + 3}$	dM1	1.1b
	Range $\frac{5}{3} \leq fg^{-1}(x) \leq \frac{37}{11}$	A1	2.1
	(3)		
(8 marks)			
Notes:			

(a)

M1: Attempts to solve $\frac{3}{2} = \frac{8x+5}{2x+3} \Rightarrow x = \dots$ You can condone poor algebra as long as they reach a value for x .

Alternatively attempt to substitute $x = \frac{3}{2}$ into $f^{-1}(x) = \frac{\pm 5 \pm 3x}{\pm 2x \pm 8}$ or equivalent (may be in terms of y). Note that attempts to find e.g. $f'(x)$ or $\frac{1}{f(x)}$ which may be implied by values such as

$\frac{6}{17}, \frac{17}{6}, \frac{7}{18}, \frac{18}{7}$ score M0

A1: Achieves $\left(f^{-1}\left(\frac{3}{2}\right)\right) = -\frac{1}{10}$. Do not be concerned what they call it, just look for the value e.g.

$x = -\frac{1}{10}$ or just $-\frac{1}{10}$ is fine. Correct answer with no (or minimal) working scores both marks.

(b)

M1: Attempts to divide $8x+5$ by $2x+3$

Look for $4 \pm \frac{\dots}{2x+3}$ where $\dots$ is a constant or $8x+5 = A(2x+3) + B$ with A or B correct

(which may be in a fraction) or in a long division attempt and obtains a quotient of 4

or attempts to express the numerator in terms of the denominator e.g. $\frac{8x+5}{2x+3} = \frac{4(2x+3) + \dots}{2x+3}$

A1: A full and complete method showing $\frac{8x+5}{2x+3} = 4 - \frac{7}{2x+3}$ or $\frac{8x+5}{2x+3} = 4 + \frac{-7}{2x+3}$

Also allow for correct values e.g. $A = 4, B = -7$

Do not isw here e.g. if they obtain $A = 4, B = -7$ and then write $-7 + \frac{4}{2x+3}$ score A0

(c)

B1: Deduces $0 \leq g^{-1}(x) \leq 4$ o.e.

E.g. $0 \leq y \leq 4, 0 \leq \text{range} \leq 4, g^{-1}(x) \leq 4$ and $g^{-1}(x) \geq 0, 0 \leq g^{-1} \leq 4, [0, 4]$

but not e.g. $0 \leq x \leq 4, 0 \leq g(x) \leq 4, (0, 4)$

(d)

M1: Attempts either boundary. Look for either $f(0) = \frac{8 \times 0 + 5}{2 \times 0 + 3}$ or $f(4) = \frac{8 \times 4 + 5}{2 \times 4 + 3}$

or uses (b) e.g. $f(0) = 4 - \frac{7}{2 \times 0 + 3}$ or $f(4) = 4 - \frac{7}{2 \times 4 + 3}$

dM1: Attempts both boundaries. Look for $f(0) = \frac{8 \times 0 + 5}{2 \times 0 + 3}$ and $f(4) = \frac{8 \times 4 + 5}{2 \times 4 + 3}$

or uses (b) e.g. $f(0) = 4 - \frac{7}{2 \times 0 + 3}$ and $f(4) = 4 - \frac{7}{2 \times 4 + 3}$

A1: Correct answer written in the correct form.

E.g. $\frac{5}{3} \leq fg^{-1}(x) \leq \frac{37}{11}, \frac{5}{3} \leq \text{range} \leq \frac{37}{11}, \frac{5}{3} \leq y \leq \frac{37}{11}, fg^{-1}(x) \leq \frac{37}{11}$ and $fg^{-1}(x) \geq \frac{5}{3}$

$\frac{5}{3} \leq fg^{-1} \leq \frac{37}{11}, fg^{-1}(x) \leq \frac{37}{11} \cap fg^{-1}(x) \geq \frac{5}{3}, \left[\frac{5}{3}, \frac{37}{11} \right]$ but not e.g. $\frac{5}{3} \leq x \leq \frac{37}{11}$

(d) **Alternative:**

M1: Attempts $fg^{-1}(x)$ and either boundary using $x = 0$ or $x = 16$

Look for either $fg^{-1}(0) = \frac{8 \times g^{-1}(0) + 5}{2 \times g^{-1}(0) + 3}$ or $fg^{-1}(16) = \frac{8 \times g^{-1}(16) + 5}{2 \times g^{-1}(16) + 3}$

Or uses (b) e.g. $fg^{-1}(0) = 4 - \frac{7}{2 \times g^{-1}(0) + 3}$ or $fg^{-1}(16) = 4 - \frac{7}{2 \times g^{-1}(16) + 3}$

The attempt at $fg^{-1}(x)$ requires an attempt to substitute $\sqrt{16-x}$ (condone $\pm \sqrt{16-x}$) into f

dM1: Attempts both boundaries. Look for $fg^{-1}(0) = \frac{8 \times g^{-1}(0) + 5}{2 \times g^{-1}(0) + 3}$ and $fg^{-1}(16) = \frac{8 \times g^{-1}(16) + 5}{2 \times g^{-1}(16) + 3}$

Or uses (b) e.g. $fg^{-1}(0) = 4 - \frac{7}{2 \times g^{-1}(0) + 3}$ and $fg^{-1}(16) = 4 - \frac{7}{2 \times g^{-1}(16) + 3}$

The attempt at $fg^{-1}(x)$ requires an attempt to substitute $\sqrt{16-x}$ (condone $\pm \sqrt{16-x}$) into f

A1: Correct answer written in the correct form with exact values.

E.g. $\frac{5}{3} \leq fg^{-1}(x) \leq \frac{37}{11}, \frac{5}{3} \leq \text{range} \leq \frac{37}{11}, \frac{5}{3} \leq y \leq \frac{37}{11}, fg^{-1}(x) \leq \frac{37}{11}$ and $fg^{-1}(x) \geq \frac{5}{3}$

$\frac{5}{3} \leq fg^{-1} \leq \frac{37}{11}, fg^{-1}(x) \leq \frac{37}{11} \cap fg^{-1}(x) \geq \frac{5}{3}, \left[\frac{5}{3}, \frac{37}{11} \right]$ but not e.g. $\frac{5}{3} \leq x \leq \frac{37}{11}$

Note that the $\frac{37}{11}$ is sometimes obtained fortuitously from incorrect working so check working carefully.

Q4.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	The asymptote is found where $2x - q = 0$ Hence $q = 4$	B1	This mark is given for explaining that the asymptote at $x = 2$ is a solution of $2x - q = 0$
	$y = \frac{p-3x}{(2x-4)(x+3)}$ $\frac{1}{2} = \frac{p-9}{(6-4)(3+3)}$	M1	This mark is given for substituting $x = 3$, $y = \frac{1}{2}$ (and $q = 4$)
(b)	$\frac{15-3x}{(2x-4)(x+3)} = \frac{A}{(2x-4)} + \frac{B}{(x+3)}$	M1	This mark is given for a method to use partial fractions
	$= \frac{1.8}{(2x-4)} - \frac{2.4}{(x+3)}$	M1	This mark is given for finding values for A and B
	$= \frac{0.9}{(x-2)} - \frac{2.4}{(x+3)}$	A1	This mark is given for a fully simplified expression
	$I = \int \frac{15-3x}{(2x-4)(x+3)} dx$ $= m \ln(2x-4) + n \ln(x+3)$	M1	This mark is given for a method to integrate to find the area of R
	$= 0.9 \ln(2x-4) + 2.4 \ln(x+3)$	A1	This mark is given for a correct expression for the area of R
	$\text{Area } R = \left[0.9 \ln(2x-4) - 2.4 \ln(x+3) \right]_3^5$	M1	This mark is given for deducing an expression for the area of R ($y = 0$ when $x = 5$)
	$= [0.9 \ln 6 - 2.4 \ln 8] - [0.9 \ln 2 - 2.4 \ln 6]$ $= [0.9 \ln 6 + 2.4 \ln 6] - [7.2 \ln 2 + 0.9 \ln 2]$ $= 3.3 \ln 6 - 8.1 \ln 2$ $= 3.3 \ln 3 + 3.3 \ln 2 - 8.1 \ln 2$	M1	This mark is given for a method to find the exact area of R
	$= 3.3 \ln 3 - 4.8 \ln 2$	A1	This mark is given for a correct value of the area of R with $a = 3.3$ and $b = 4.8$

Q5.

Question	Scheme	Marks	AOs
(a)	$h = 0.5$	B1	1.1a
	$A \approx \frac{0.5}{2} \{0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)\}$	M1	1.1b
	= awrt 1.50	A1	1.1b
	For reference: The integration on a calculator gives 1.511549071 The full accuracy for y values gives 1.504726147 The accuracy from the table gives 1.50475		
		(3)	
(b)	$3 \times \text{their (a)}$ If (a) is correct, allow awrt 4.50 or awrt 4.51 even with no working. Only allow 4.5 if (a) is correct and working is shown e.g. 3×1.5 If (a) is incorrect allow $3 \times \text{their (a)}$ given to at least 3sf but do not be too concerned about the accuracy (as they may use rounded or rounded value from (a))	B1ft	2.2a
	For reference the integration on a calculator gives 4.534647213		
		(1)	
(c)	<u>This mark depends on the B1 having been awarded in part (b) with awrt 4.5</u> Look for a sensible comment. Some examples: - The answer is accurate to 2 sf or one decimal place - Answer to (b) is accurate as $4.535 \approx 4.50$ - Very accurate as 4.535 to 2 sf is 4.5 $4.51425 < 4.535$ so my answer is underestimate but not too far off - It is an underestimate but quite close - It is a very good estimate - High accuracy (Quite) accurate - It is less than 1% out $4.535 - 4.5 = 0.035$ so not far out But not just "it is an underestimate" or Calculates percentage error correctly using awrt 4.50 or awrt 4.51 or 4.5 (No comment is necessary in these cases although one may be given) Examples: $\left \frac{4.535 - 4.50}{4.535} \right \times 100 = 0.77\% \quad \text{or} \quad \left \frac{4.535 - 4.51}{4.535} \right \times 100 = 0.55\% \quad \text{or}$ $\left \frac{4.535 - 4.51425}{4.535} \right \times 100 = 0.46\% \quad \text{or} \quad \left \frac{4.50}{4.535} \right \times 100 = 99\%$ In these cases don't be too concerned about accuracy e.g. allow 1sf. This mark should be withheld if there are any contradictory statements	B1	3.2b
		(1)	
(5 marks)			

Notes:

(a)

B1: States or uses $h = 0.5$. May be implied by $\frac{1}{4} \times \{ \dots \}$ below.

M1: Correct attempt at the trapezium rule.

Look for $\frac{1}{2}h \times \{0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)\}$ condoning slips on the terms but must use all y values with no repeats.

There must be a clear attempt at $\frac{1}{2}h \times (\text{first } y + \text{last } y + 2 \times \text{"sum of the rest"})$

Give M0 for $\frac{1}{2} \times \frac{1}{2} \times 0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)$ unless the missing brackets are implied.

NB this incorrect method gives 5.85...

May be awarded for separate trapezia e.g.

$$\frac{1}{4}(0.5774 + 0.7071) + \frac{1}{4}(0.7071 + 0.7746) + \frac{1}{4}(0.7746 + 0.8165) + \frac{1}{4}(0.8165 + 0.8452)$$

May be awarded for using the function e.g. $\frac{1}{2}h \times \left\{ \sqrt{\frac{0.5}{1+0.5}} + \sqrt{\frac{2.5}{1+2.5}} + 2 \left(\sqrt{\frac{1}{1+1}} + \sqrt{\frac{1.5}{1+1.5}} + \sqrt{\frac{2}{1+2}} \right) \right\}$

A1: Awrt 1.50 (Apply isw if necessary)

Correct answers with no working – send to review

(b)

B1ft: See main scheme. Must be considering $3 \times (a)$ and not e.g. attempting trapezium rule again.

(c)

B1: See scheme

Q6.

Question	Scheme	Marks	AOs
(a)(i)	$50x^2 + 38x + 9 \equiv A(5x+2)(1-2x) + B(1-2x) + C(5x+2)^2$ $\Rightarrow B = \dots$ or $C = \dots$	M1	1.1b
	$B = 1$ and $C = 2$	A1	1.1b
(a)(ii)	E.g. $x = 0$ $x = 0 \Rightarrow 9 = 2A + B + 4C$ $\Rightarrow 9 = 2A + 1 + 8 \Rightarrow A = \dots$	M1	2.1
	$A = 0^*$ Q6.	A1*	1.1b
		(4)	
(b)(i)	$\frac{1}{(5x+2)^2} = (5x+2)^{-2} = 2^{-2} \left(1 + \frac{5}{2}x\right)^{-2}$ or $(5x+2)^{-2} = 2^{-2} + \dots$	M1	3.1a
	$\left(1 + \frac{5}{2}x\right)^{-2} = 1 - 2\left(\frac{5}{2}x\right) + \frac{-2(-2-1)}{2!}\left(\frac{5}{2}x\right)^2 + \dots$	M1	1.1b
	$2^{-2} \left(1 + \frac{5}{2}x\right)^{-2} = \frac{1}{4} - \frac{5}{4}x + \frac{75}{16}x^2 + \dots$	A1	1.1b
	$\frac{1}{(1-2x)} = (1-2x)^{-1} = 1 + 2x + \frac{-1(-1-1)}{2!}(2x)^2 + \dots$	M1	1.1b
	$\frac{1}{(5x+2)^2} + \frac{2}{1-2x} = \frac{1}{4} - \frac{5}{4}x + \frac{75}{16}x^2 + \dots + 2 + 4x + 8x^2 + \dots$	dM1	2.1
	$= \frac{9}{4} + \frac{11}{4}x + \frac{203}{16}x^2 + \dots$	A1	1.1b
(b)(ii)	$ x < \frac{2}{5}$	B1	2.2a
		(7)	
(11 marks)			
Notes			

(a)(i)

M1: Uses a correct identity and makes progress using an appropriate strategy (e.g. sub $x = \frac{1}{2}$) to find a value for B or C . May be implied by one correct value (cover up rule).

A1: Both values correct

(a)(ii)

M1: Uses an appropriate method to establish an equation connecting A with B and/or C and uses their values of B and/or C to find a suitable equation in A .

Amongst many different methods are:

Compare terms in $x^2 \Rightarrow 50 = -10A + 25C$ which would be implied by $50 = -10A + 25 \times 2$

Compare constant terms or substitute $x = 0 \Rightarrow 9 = 2A + B + 4C$ implied by $9 = 2A + 1 + 4 \times 2$

A1*: Fully correct proof with no errors.

Note: The second part is a proof so it is important that a suitable proof/show that is seen.

Candidates who write down 3 equations followed by three answers (with no working) will score M1 A1 M0 A0

(b)(i)

M1: Applies the key steps of writing $\frac{1}{(5x+2)^2}$ as $(5x+2)^{-2}$ and takes out a factor of 2^{-2} to form an

expression of the form $(5x+2)^{-2} = 2^{-2}(1+*x)^{-2}$ where * is not 1 or 5

Alternatively uses direct expansion to obtain $2^{-2} + \dots$

M1: Correct attempt at the binomial expansion of $(1+*x)^{-2}$ up to the term in x^2

Look for $1+(-2)*x + \frac{(-2)(-3)}{2}*x^2$ where * is not 5 or 1.

Condone sign slips and lack of $*^2$ on term 3.

Alt Look for correct structure for 2nd and 3rd terms by direct expansion. See below

A1: For a fully correct expansion of $(2+5x)^{-2}$ which may be unsimplified. This may have been combined with their 'B'

A direct expansion would look like $(2+5x)^{-2} = 2^{-2} + (-2)2^{-3} \times 5x + \frac{(-2)(-3)}{2}2^{-4} \times (5x)^2$

M1: Correct attempt at the binomial expansion of $(1-2x)^{-1}$

Look for $1+(-1)*x + \frac{(-1)(-2)}{2}*x^2$ where * is not 1

dM1: Fully correct strategy that is dependent on the previous **TWO** method marks.

There must be some attempt to use their values of B and C

A1: Correct expression or correct values for p, q and r.

(b)(ii)

B1: Correct range. Allow also other forms, for example $-\frac{2}{5} < x < \frac{2}{5}$ or $x \in \left(-\frac{2}{5}, \frac{2}{5}\right)$

Do not allow multiple answers here. The correct answer must be chosen if two answers are offered

Q7.

Question	Scheme	Marks	AOs
(a)	$f(x) = 4(x^2 - 2)e^{-2x}$		
	Differentiates to $e^{-2x} \times 8x + 4(x^2 - 2) \times -2e^{-2x}$	M1 A1	1.1b 1.1b
	$f'(x) = 8e^{-2x} \{x - (x^2 - 2)\} = 8(2 + x - x^2)e^{-2x}$ *	A1*	2.1
		(3)	
(b)	States roots of $f'(x) = 0$ $x = -1, 2$	B1	1.1b
	Substitutes one x value to find a y value	M1	1.1b
	Stationary points are $(-1, -4e^2)$ and $(2, 8e^{-4})$	A1	1.1b
		(3)	
(c)	(i) Range $[-8e^2, \infty)$ o.e. such as $g(x) \geq -8e^2$	B1ft	2.5
	(ii) For - Either attempting to find $2f(0) - 3 = 2 \times -8 - 3 = (-19)$ and identifying this as the lower bound - Or attempting to find $2 \times 8e^{-4} - 3$ and identifying this as the upper bound	M1	3.1a
	Range $[-19, 16e^{-4} - 3]$	A1	1.1b
		(3)	
(9 marks)			
Notes:			

(a)

M1: Attempts the product rule and uses $e^{-2x} \rightarrow ke^{-2x}$, $k \neq 0$

If candidate states $u = 4(x^2 - 2)$, $v = e^{-2x}$ with $u' = \dots$, $v' = \dots e^{-2x}$ it can be implied by their $vu' + uv'$

If they just write down an answer without working award for $f'(x) = pxe^{-2x} \pm q(x^2 - 2)e^{-2x}$

They may multiply out first $f(x) = 4x^2e^{-2x} - 8e^{-2x}$. Apply in the same way condoning slips

Alternatively attempts the quotient rule on $f(x) = \frac{u}{v} = \frac{4(x^2 - 2)}{e^{2x}}$ with $v' = ke^{2x}$ and $f'(x) = \frac{vu' - uv'}{v^2}$

A1: A correct $f'(x)$ which may be unsimplified.

Via the quotient rule you can award for $f'(x) = \frac{8xe^{2x} - 8(x^2 - 2)e^{2x}}{e^{4x}}$ o.e.

A1*: Proceeds correctly to given answer showing all necessary steps.

The $f'(x)$ or $\frac{dy}{dx}$ must be present at some point in the solution

This is a "show that" question and there must not be any errors. All bracketing must be correct.

Allow a candidate to move from the **simplified** unfactorised answer of $f'(x) = 8xe^{-2x} - 8(x^2 - 2)e^{-2x}$

to the given answer in one step.

Do not allow it from an **unsimplified** $f'(x) = 4 \times 2xe^{-2x} + 4(x^2 - 2) \times -2e^{-2x}$

Allow the expression / bracketed expression to be written in a different order.

So, for example, $8(x - x^2 + 2)e^{-2x}$ is OK

(b)

B1: States or implies $x = -1, 2$ (as the roots of $f'(x) = 0$)

M1: Substitutes one x value of their solution to $f'(x) = 0$ in $f(x)$ to find a y value.

Allow decimals here (3sf). FYI, to 3 sf, $-4e^2 = -29.6$ and $8e^{-4} = 0.147$

Some candidates just write down the x coordinates but then go on in part (c) to find the ranges using the y coordinates. Allow this mark to be scored from work in part (c)

A1: Obtains $(-1, -4e^2)$ and $(2, 8e^{-4})$ as the stationary points. This must be scored in (b). Remember to isw

after a correct answer. Allow these to be written separately. E.g. $x = -1, y = -4e^2$

Extra solutions, e.g. from $x = 0$ will be penalised on this mark.

(c)(i)

B1ft: For a correct range written using correct notation.

Follow through on $2 \times$ their minimum "y" value from part (b), providing it is negative.

Condone a decimal answer if this is consistent with their answer in (b) to 3sf or better.

Examples of correct responses are $[-8e^2, \infty)$, $g \geq -8e^2$, $y \geq -8e^2$, $\{q \in \mathbb{R}, q \geq -8e^2\}$

(c)(ii)

M1: See main scheme. Follow through on $2 \times$ their " $8e^{-4}$ " $- 3$ for the upper bound.

A1: Range $[-19, 16e^{-4} - 3]$ o.e. such as $-19 \leq y \leq 16e^{-4} - 3$ but must be exact